GRAVITATION

Important formulas and concepts directly fom the NCERT-

❖ **Kepler's Laws**-

1) **Law of orbits –** "All planets move in elliptical orbits with the sun situated at one of the foci".
2) **Law of areas** – "the line that joins any planet to the sun sweeps equal areas in equal intervals of time".

$$\Delta A/\Delta t = \text{constant}$$

Planets appear to move slower when they are farther from the sun.

3) **Law of periods –**"The square of time period of revolution of a planet is proportional to the cube of the semi-major axis of the ellipse traced out by planet.

$$T^2 = (4\Pi^2/GM_s)R^3$$

M_s = mass of sun

❖ **Universal Law of Gravitation**-

➢ Every body in the universe attracts every other with a Force-**Gravitational Force.**

$$F_g = Gm_1m_2/r^2$$

This force is inversely prop. To square of distance b/w them.

➢ This law refers to **point masses** whereas we deal with extended objects which have finite size.

➢ In a collection of point masses, the force on any one of them is the vector sum of gravitational forces exerted by the other point masses.

➢ **The force of attraction b/w a hollow spherical shell of uniform density and a point mass situated outside is just as if entire mass of the shell is concentrated at centre of shell.**

➢ **The force of attraction due to a hollow spherical shell of uniform density , on a point mass situated inside is zero.**

➢ Gravitational force is **conservative** and **central.**

❖ **Gravitational constant –**

➢ Value = **6.67×10^{-11} Nm²/kg²**

❖ **Acceleration due to Gravity(g)-**

➢ **$g = GM_E/R^2_E$**

➢ **$g = 4/3\ \Pi\rho gR$**

➢ **$g = 9.8\ m/s^2$**

❖ Acceleration due to Gravity below and above the surface of earth-

- **At height h,**

$$g(h) = GM_E/(R_E+h)^2 = gR^2_E/(R_E+h)^2$$ **[g=acceleration due to gravity at surface]**

  - **For $h<R_E$,**

$$g(h) = g(1-2h/R_E)$$

- **At depth d,**

$$g(d) = g(1-d/R)$$

➢ **"acceleration due to gravity maximum at surface decreasing both at above and below the surface".**

❖ Gravitational Potential energy-

➢ The change in potential energy is just the amount of work done on the body by the force.

➢ Potential anergy at a point at a height h above the surface,

$$W(h) = mgh + Wo$$

Wo = constant (potential energy at surface of earth)

➢ The gravitational potential energy associated with two particles of masses m_1 and m_2 separated by distance r,

$$U = - Gm_1m_2/r$$

➢ Total potential energy equals to sum of energies for all possible pairs of constituent particles.

❖ Escape Speed-

➢ The minimum speed required for an object to reach infinity (i.e. escape from earth) is called "escape speed".

$$V_{min} = (2GM_E/R_E)^{1/2}$$
$$V_{min} = (2gR_E)^{1/2}$$

Vmin = 11.2 km/s

➢ **Moon** has escape speed 2.3km/s , about five times smaller than earth.
That's why moon has no atmosphere , gas molecules if formed on the surface of moon having velocities larger than this esacpe velocity will escape the gravitational pull of moon.

❖ Earth Satellites-

➢ Earth satellites are the objects which revolve around the earth.

➢ Motion is similar to planets hence, Kepler's laws are equally applicable to them.

- **Moon is the only natural satellite of earth with a near circular orbit with a time period of approx. 27.3 days which also roughly equal to the rotational period of the moon about its own axis.**

- **Time Period of motion of satellite which is very close to the surface of earth ($h<R_E$),**

$$T = 2\pi(R_E/g)1/2$$

If $g = 9.8 m/s^2$, $R_E = 6400km$
Then T is approx. 85 minutes.

❖ Energy of an orbiting satellite-

- Kinetic energy of the satellite in a circular orbit with speed υ is,

$$K.E. = \frac{1}{2} m\upsilon^2 = GmM_E/2(R_E+h)$$

- Potential energy at distance (R+h) from centre of the earth is, (considering potential energy at infinity is zero)

$$P.E. = - GmM_E/(R_E+h)$$

- Total energy ,

$$E = K.E. + P.E.$$
$$E = - GmM_E / 2(R_E+h)$$

- **Total energy is negative , if total energy is zero or positive , the object escapes to infinity.**
- **Satellites are always at finite distance from the earth and hence their energies can't be positive or zero.**

❖ Geostationary and Polar satellite-

○ **Geostationary satellite** –

- Satellites in a circular orbits around the earth in the equatorial plane with **T = 24 hours .**
- **Height from surface = 35800km, much larger thamn R_E.**
- **Since earth rotates with same period , the satellite appear stationary from any point on earth.**
- The **INSAT** group of satellites sent up by India are one such group of geostationary satellites widely used for **telecommunications** in India.

○ **Polar satellite-**

- **Low altitude (h=500-800km) satellites.**
- **They go around the poles of the earth in a north-south direction.**
- **Time period is around 100 minutes.**
- **Useful for remote sensing, meterology as well as for envitonmental studies of earth.**

- **Weightlessness-**
  - **When an object is in free fall , it is weightless and this phenomenon is weightlessness.**
  - **In a manned satellite people inside experience no gravity.**

SUMMARY

1. Newton's law of universal gravitation states that the gravitational force of attraction between any two particles of masses m_1 and m_2 separated by a distance r has the magnitude

$$F = G\frac{m_1 m_2}{r^2}$$

where G is the universal gravitational constant, which has the value 6.672×10^{-11} N m^2 kg^{-2}.

2. If we have to find the resultant gravitational force acting on the particle m due to a number of masses $M_1, M_2, \ldots.M_n$ etc. we use the principle of superposition. Let $F_1, F_2, \ldots.F_n$ be the individual forces due to $M_1, M_2, \ldots.M_n$, each given by the law of gravitation. From the principle of superposition each force acts independently and uninfluenced by the other bodies. The resultant force F_R is then found by vector addition

$$F_R = F_1 + F_2 + \ldots\ldots + F_n = \sum_{i=1}^{n} F_i$$

where the symbol 'Σ' stands for summation.

3. Kepler's laws of planetary motion state that
   (a) All planets move in elliptical orbits with the Sun at one of the focal points
   (b) The radius vector drawn from the Sun to a planet sweeps out equal areas in equal time intervals. This follows from the fact that the force of gravitation on the planet is central and hence angular momentum is conserved.
   (c) The square of the orbital period of a planet is proportional to the cube of the semi-major axis of the elliptical orbit of the planet

   The period T and radius R of the circular orbit of a planet about the Sun are related by

$$T^2 = \left(\frac{4\pi^2}{G\,M_s}\right)R^3$$

where M_s is the mass of the Sun. Most planets have nearly circular orbits about the Sun. For elliptical orbits, the above equation is valid if R is replaced by the semi-major axis, a.

4. The acceleration due to gravity.
   (a) at a height h above the earth's surface

$$g(h) = \frac{G\,M_E}{(R_E + h)^2}$$

$$\approx \frac{G\,M_E}{R_E^2}\left(1 - \frac{2h}{R_E}\right) \quad \text{for } h << R_E$$

$$g(h) = g(0)\left(1 - \frac{2h}{R_E}\right) \quad \text{where } g(0) = \frac{G\,M_E}{R_E^2}$$

(b) at depth d below the earth's surface is

$$g(d) = \frac{G\,M_E}{R_E^2}\left(1 - \frac{d}{R_E}\right) = g(0)\left(1 - \frac{d}{R_E}\right)$$

5. The gravitational force is a conservative force, and therefore a potential energy function can be defined. The *gravitational potential energy* associated with two particles separated by a distance r is given by

$$V = -\frac{G\,m_1\,m_2}{r}$$

where V is taken to be zero at $r \to \infty$. The total potential energy for a system of particles is the sum of energies for all pairs of particles, with each pair represented by a term of the form given by above equation. This prescription follows from the principle of superposition.

6. If an isolated system consists of a particle of mass m moving with a speed v in the vicinity of a massive body of mass M, the total mechanical energy of the particle is given by

$$E = \frac{1}{2}m\,v^2 - \frac{G\,M\,m}{r}$$

That is, the total mechanical energy is the sum of the kinetic and potential energies. The total energy is a constant of motion.

7. If m moves in a circular orbit of radius a about M, where $M >> m$, the total energy of the system is

$$E = -\frac{G\,M\,m}{2a}$$

with the choice of the arbitrary constant in the potential energy given in the point 5., above. The total energy is negative for any bound system, that is, one in which the orbit is closed, such as an elliptical orbit. The kinetic and potential energies are

$$K = \frac{G\,M\,m}{2a}$$

$$V = -\frac{G\,M\,m}{a}$$

8. The escape speed from the surface of the earth is

$$v_e = \sqrt{\frac{2\,G\,M_E}{R_E}} = \sqrt{2gR_E}$$

and has a value of 11.2 km s^{-1}.

9. If a particle is outside a uniform spherical shell or solid sphere with a spherically symmetric internal mass distribution, the sphere attracts the particle as though the mass of the sphere or shell were concentrated at the centre of the sphere.

10. If a particle is inside a uniform spherical shell, the gravitational force on the particle is zero. If a particle is inside a homogeneous solid sphere, the force on the particle acts toward the centre of the sphere. This force is exerted by the spherical mass interior to the particle.

Important points from NCERT summary-

- **A geostationary satellite moves at an approximate distance of 4.22×10^4 km from earth's surface.**

Note – for this chapter , points to ponder are very very important. Don't skip any point.

POINTS TO PONDER

1. In considering motion of an object under the gravitational influence of another object the following quantities are conserved:
   (a) Angular momentum
   (b) Total mechanical energy
   Linear momentum is **not** conserved
2. Angular momentum conservation leads to Kepler's second law. However, it is not special to the inverse square law of gravitation. It holds for any central force.
3. In Kepler's third law (see Eq. (7.1) and $T^2 = K_S R^3$. The constant K_S is the same for all planets in circular orbits. This applies to satellites orbiting the Earth [(Eq. (7.38)].
4. An astronaut experiences weightlessness in a space satellite. This is not because the gravitational force is small at that location in space. It is because both the astronaut and the satellite are in "free fall" towards the Earth.
5. The *gravitational potential energy* associated with two particles separated by a distance r is given by

$$V = -\frac{G m_1 m_2}{r} + \text{constant}$$

   The constant can be given any value. The simplest choice is to take it to be zero. With this choice

$$V = -\frac{G m_1 m_2}{r}$$

   This choice implies that $V \to 0$ as $r \to \infty$. Choosing location of zero of the gravitational energy is the same as choosing the arbitrary constant in the potential energy. Note that the gravitational force is not altered by the choice of this constant.
6. The total mechanical energy of an object is the sum of its kinetic energy (which is always positive) and the potential energy. Relative to infinity (i.e. if we presume that the potential energy of the object at infinity is zero), the gravitational potential energy of an object is negative. The total energy of a satellite is negative.
7. The commonly encountered expression mgh for the potential energy is actually an approximation to the difference in the gravitational potential energy discussed in the point 6, above.
8. Although the gravitational force between two particles is central, the force between two finite rigid bodies is not necessarily along the line joining their centre of mass. For a spherically symmetric body however the force on a particle external to the body is as if the mass is concentrated at the centre and this force is therefore central.
9. The gravitational force on a particle inside a spherical shell is zero. However, (unlike a metallic shell which shields electrical forces) the shell does not shield other bodies outside it from exerting gravitational forces on a particle inside. *Gravitational shielding is not possible.*